

Functions

1. Functions

1.1. Definition

- A function is a relation from a set X to a set Y where every element $x \in X$ is mapped to exactly one element $y \in Y$.
- Multiple inputs can map to the same output, but a single input can never map to multiple outputs. As a helpful reminder, consider $f(x) = x^2$: both inputs -2 and 2 map to the same output 4 .
- Vertical Line Test: Determines whether a given graph represents a function. Draw vertical lines across the graph; if any vertical line intersects the graph more than once, it is not a function.

1.2. Domain, Codomain and Range

- Domain: The set of all possible input values that produces real output values.
- Codomain: The set of all possible theoretical output values a function can produce.
- Range: The set of actual output values a function produces.

1.3. Finding the Domain

- To find the domain we find the restricted input values, these are the input values that do not produce any real value.

Pattern	Restrictions	Example
With fractions	Denominator $\neq 0$	$f(x) = \frac{1}{x-3}$ $\Rightarrow x-3 \neq 0$ $\Rightarrow x \neq 3$ $\therefore$ The domain of $f(x)$ is $(-\infty, 3) \cup (3, \infty)$
With even roots $\sqrt{}$, $\sqrt[2]{}$, $\sqrt[4]{}$	Radicand ≥ 0	$f(x) = \sqrt{x-3}$ $\Rightarrow x-3 \geq 0$ $\Rightarrow x \geq 3$ $\therefore$ The domain of $f(x)$ is $[3, \infty)$
With logarithms $\log_b()$, $\ln()$	Argument > 0	$f(x) = \ln(x-5)$ $\Rightarrow x-5 > 0$ $\Rightarrow x > 5$ $\therefore$ The domain of $f(x)$ is $(5, \infty)$

1.4. Function Operations

- $(f+g)(x) = f(x) + g(x)$
- $(f-g)(x) = f(x) - g(x)$
- $(f \cdot g)(x) = f(x) \cdot g(x)$
- $(f/g)(x) = \frac{f(x)}{g(x)}$

1.5. Composite Functions

- $(f \circ g)(x) = f(g(x))$
- Domain of $(f \circ g)$: All x in the domain of g such that $g(x)$ is in the domain of f .

1.6. Piecewise Functions

- Functions defined by different rules over different sub-domains.

Ex.

$$f(x) = \begin{cases} x + 2 & \text{if } x < 0 \\ x^2 & \text{if } x \geq 0 \end{cases}$$

1.7. One to One (Injective) Function

- A function is one to one if every input in the domain produces a unique output.
- A function is one to one if and only if $f(a) = f(b)$ implies that $a = b$.
- Horizontal Line Test: Draw horizontal lines across different heights of the functions graph. If any horizontal line intersects the graph more than once, it's not a one to one function and if every Horizontal line intersects the graph at exactly one point then it is a one to one function.

1.8. Onto (Surjective) Function

- A function is onto if the codomain of the function matches the range of a function (Every potential output hit).

1.9. Bijective Function

- Both One to One and Onto

1.10. Inverse Function

- An inverse function is a function that “reverses” or “undoes” the operation of the corresponding function.
- Two-Sided Inverse: A function $f^{\{-1\}}$ is a true two-sided inverse of f if it undoes f regardless of the order of composition:

$$(f^{-1} \circ f)(x) = x \quad \text{for all } x \in X$$

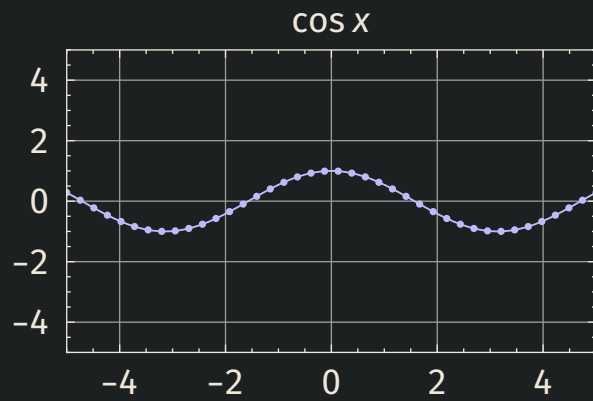
$$(f \circ f^{-1})(y) = y \quad \text{for all } y \in Y$$

- A function has a two-sided inverse if and only if it is bijective (both one-to-one and onto):
 - Left Inverse only ($f^{-1} \circ f = \text{id}$): Guarantees the function is injective (one-to-one).
 - Right Inverse only ($f \circ f^{-1} = \text{id}$): Guarantees the function is surjective (onto).
- A function and its inverse function share the same points, but with swapped coordinates. So, $(-2, 1)$ becomes $(1, -2)$.
- The domain of $f(x)$ becomes the range of $f^{-1}(x)$ and the range of $f(x)$ becomes the domain of $f^{-1}(x)$.
- The graph of $f^{-1}(x)$ is a reflection of the graph of $f(x)$ across the line $y = x$.
- To find the inverse function algebraically:
 1. Replace $f(x)$ with y .
 2. Swap x and y in the equation.
 3. Solve for y in the new equation.
 4. Replace y with $f^{-1}(x)$.

1.11. Even/Odd

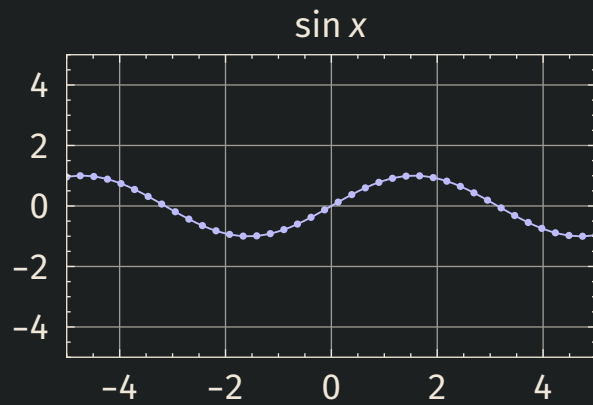
- Even Functions: Symmetric about the y -axis (Mirror image about the y -axis).

$$f(-x) = f(x)$$



- Odd Functions: Symmetric about the origin (Remains same when rotated 180° across the origin).

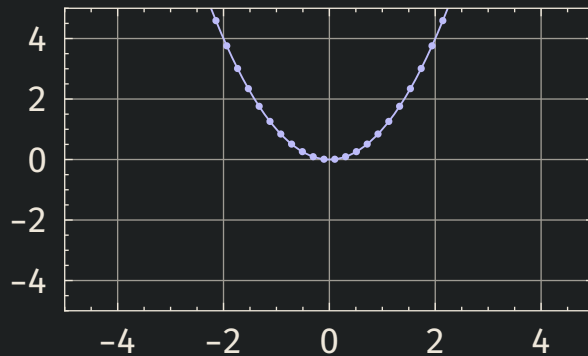
$$f(-x) = -f(x)$$



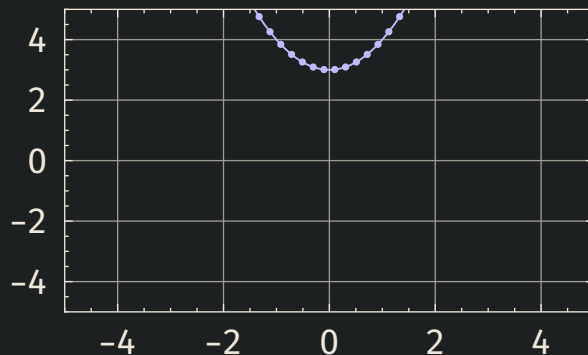
2. Transformation of functions

2.1. Translations

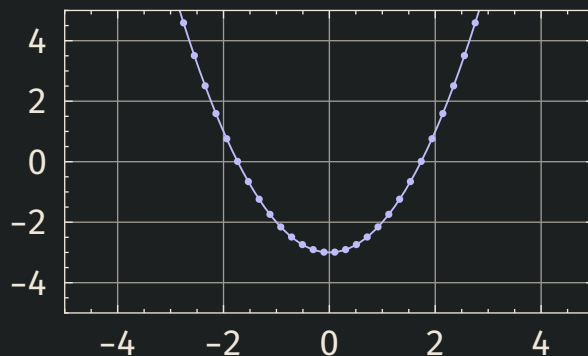
- $f(x) = x^2 \rightarrow$ Parent Function
 x^2



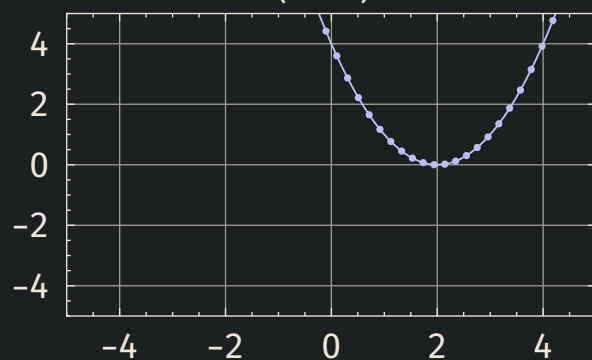
- $f(x) + c \rightarrow$ Shifts the graph c units up:
 $x^2 + 3$



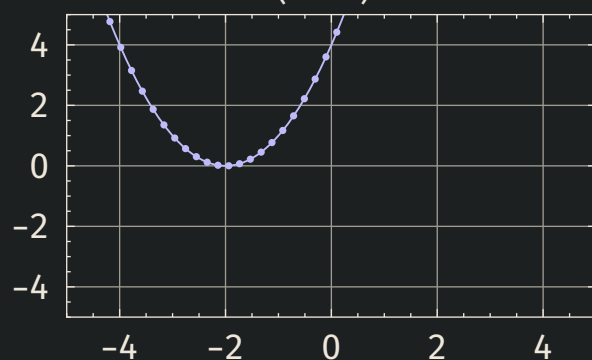
- $f(x) - c \rightarrow$ Shifts the graph c units down
 $x^2 - 3$



- $f(x - c) \rightarrow$ Shifts the graph c units right
 $(x - 2)^2$

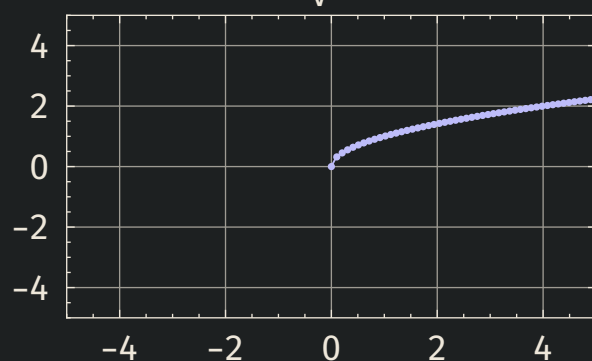


- $f(x + c) \rightarrow$ Shifts the graph c units left
 $(x + 2)^2$

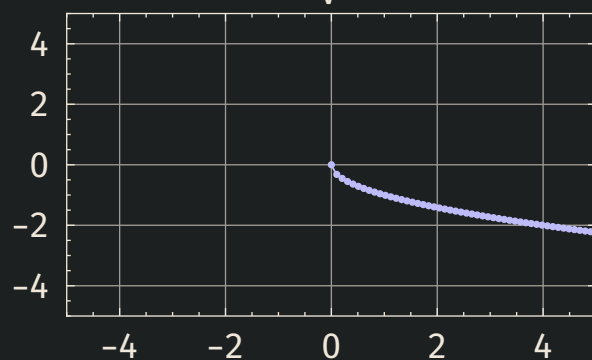


2.2. Reflections

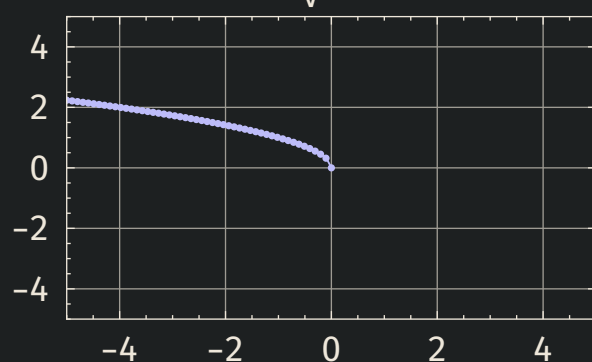
- $f(x) = \sqrt{x} \rightarrow$ Parent Function
 $\sqrt{x}$



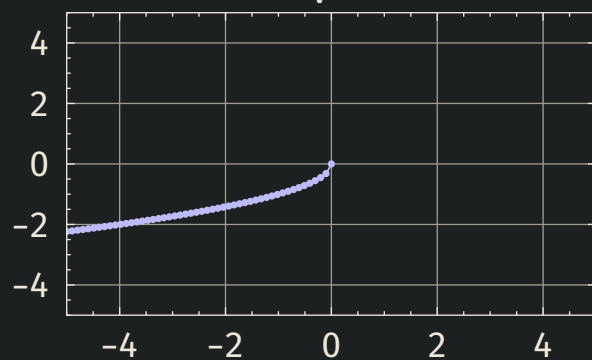
- $-f(x) \rightarrow$ Flips graph over the x-axis
 $-\sqrt{x}$



- $f(-x) \rightarrow$ Flips graph over the y-axis
 $\sqrt{-x}$

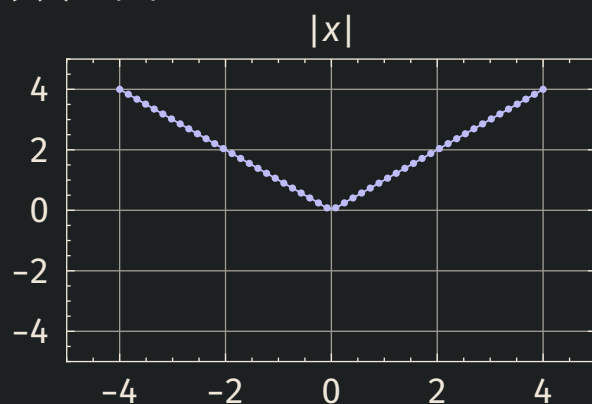


- $-f(-x) \rightarrow$ 180° rotation over the origin
 $-\sqrt{-x}$

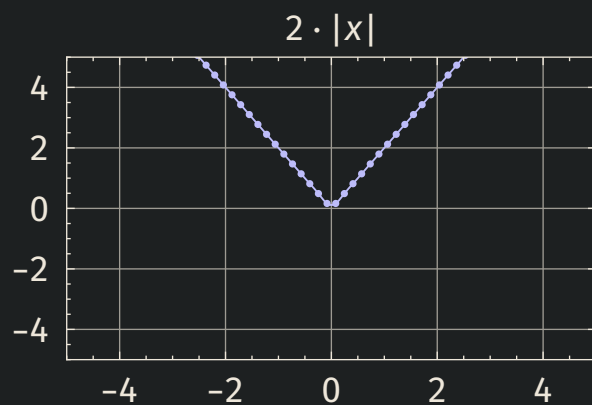


2.3. Dilations

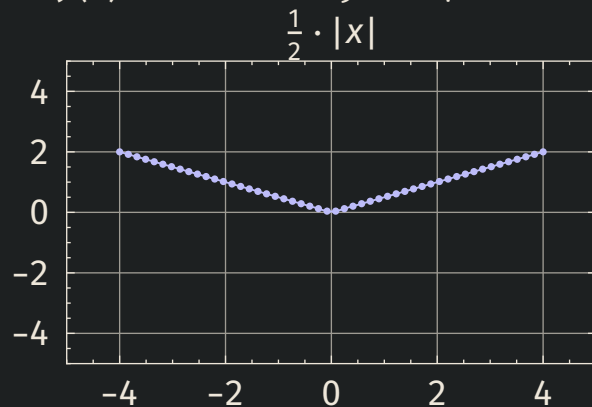
- $f(x) = |x| \rightarrow$ Parent Function



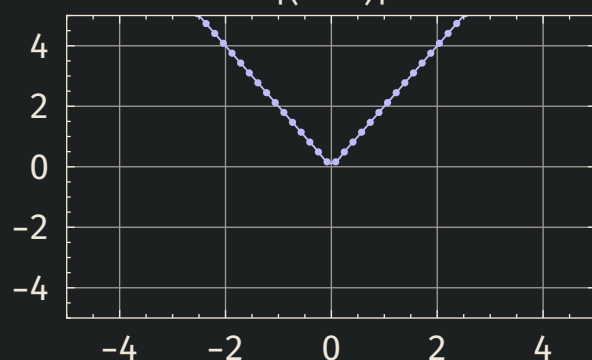
- $a \cdot f(x) \rightarrow a > 1$, Graph is stretched vertically



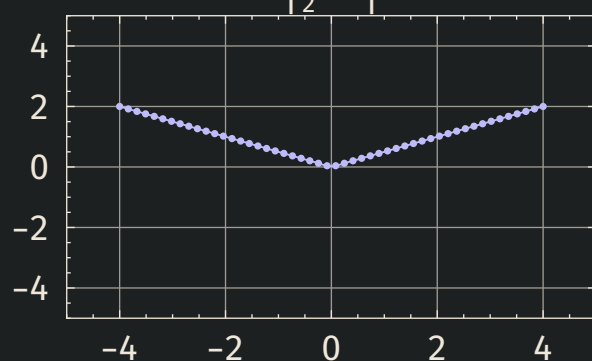
- $a \cdot f(x) \rightarrow 0 < a < 1$, Graph is shrunk vertically



- $f(a \cdot x) \rightarrow a > 1$, Graph is shrunk horizontally
 $|2 \cdot x|$

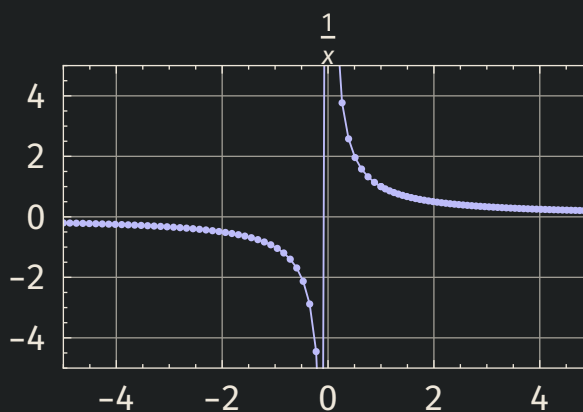
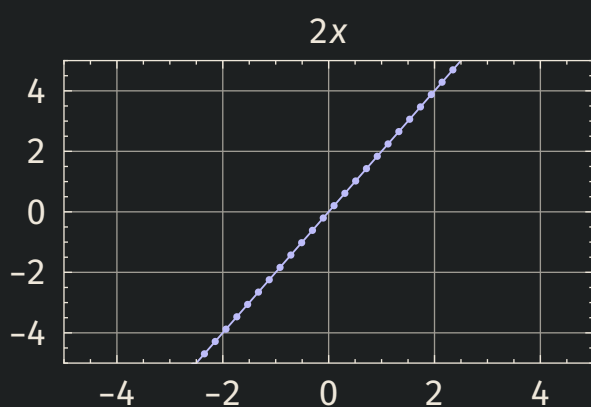


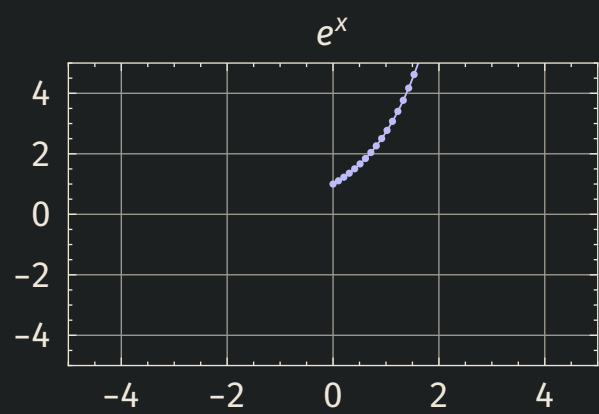
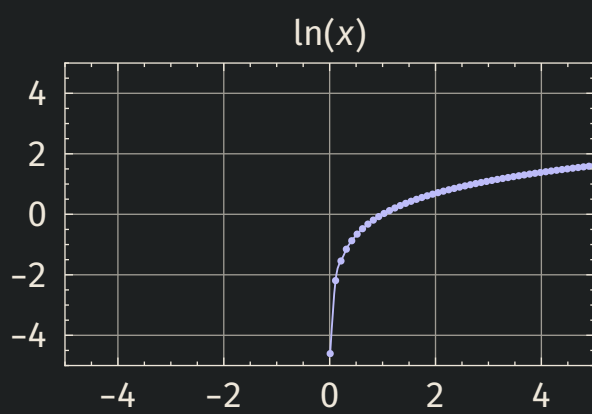
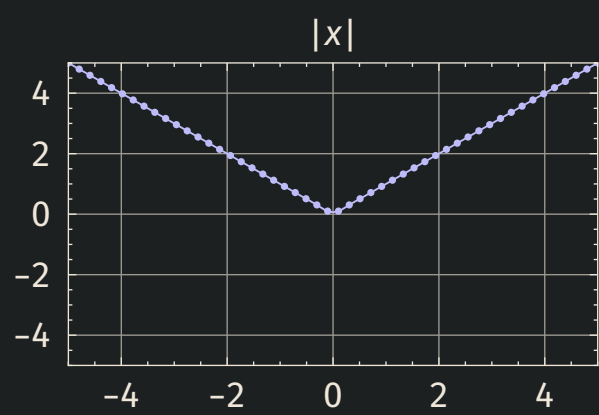
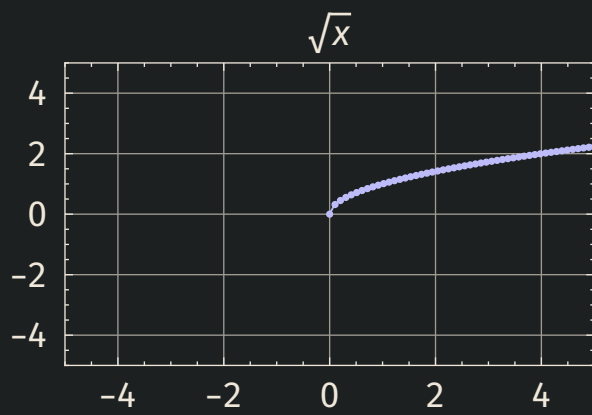
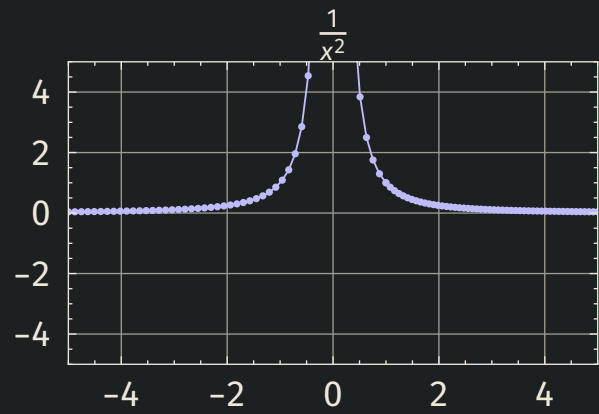
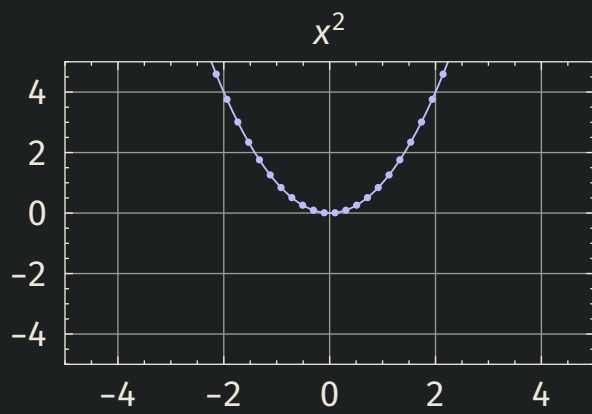
- $f(a \cdot x) \rightarrow 0 < a < 1$, Graph is stretched horizontally
 $|\frac{1}{2} \cdot x|$



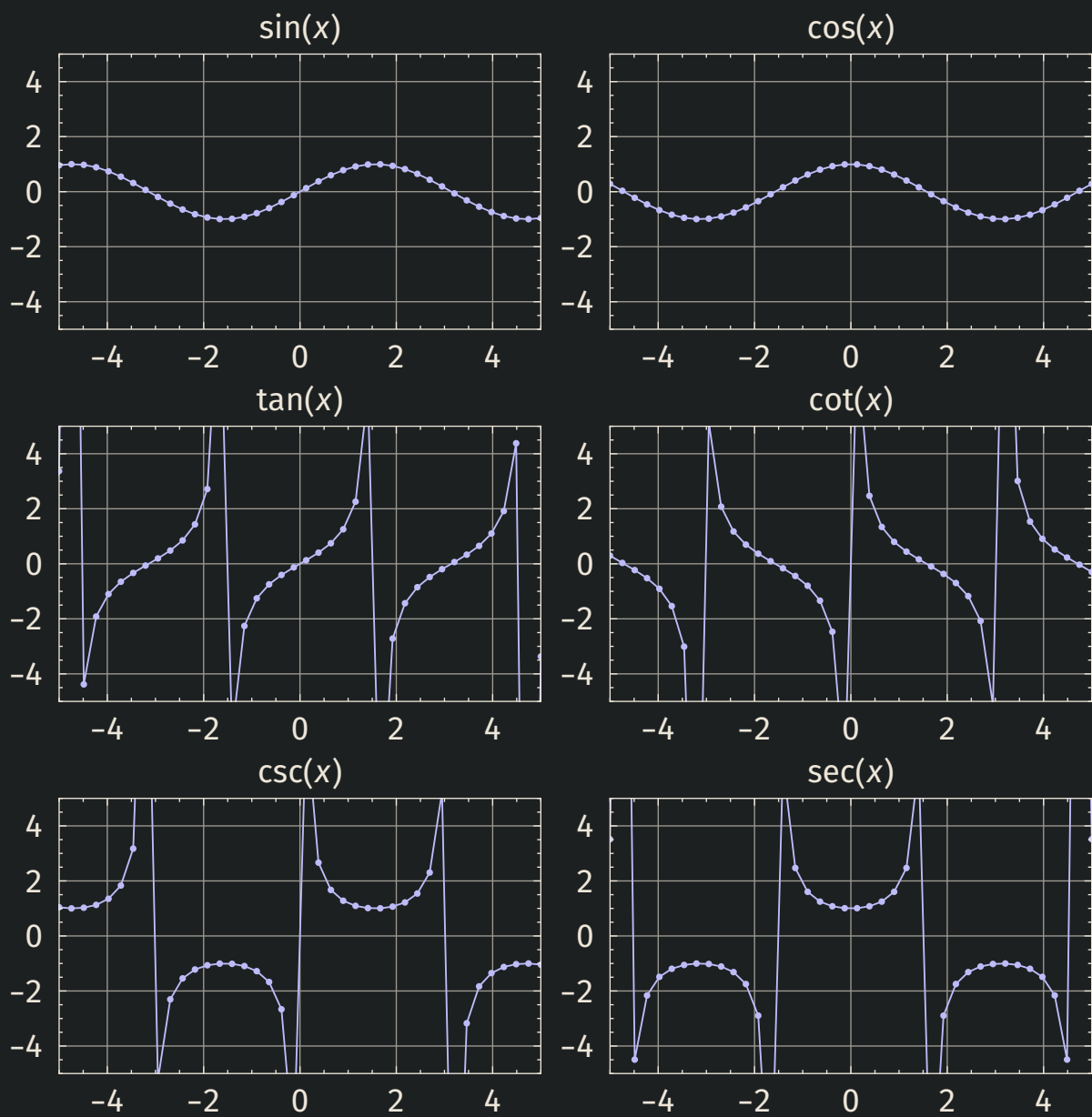
3. Graphs

3.1. Common Graphs

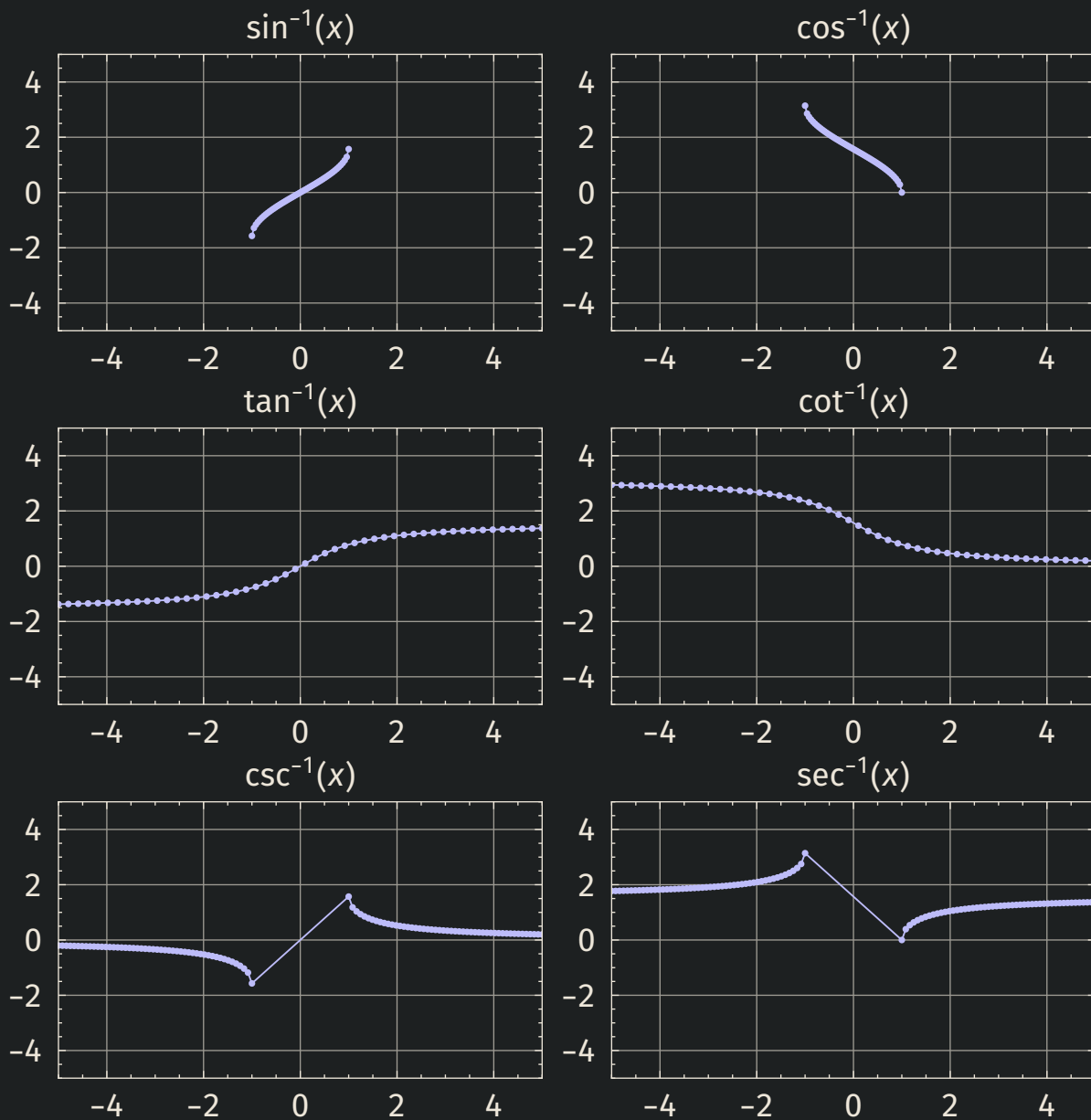




3.2. Trig Functions

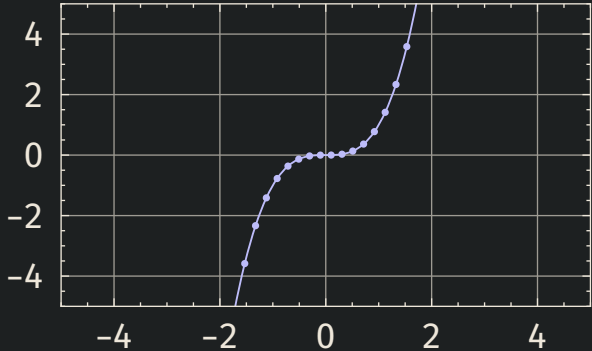
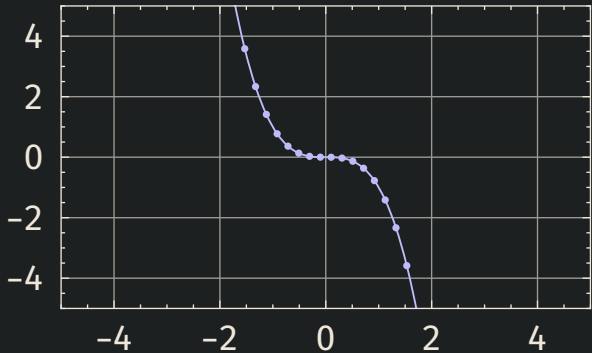
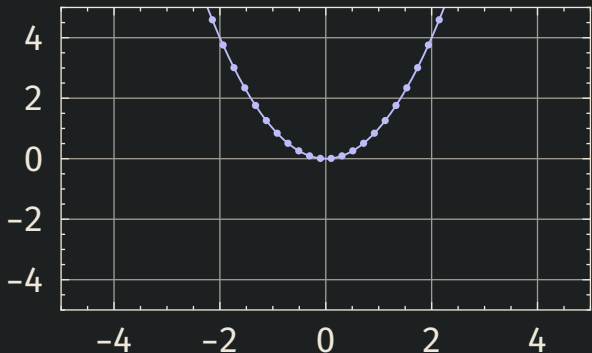


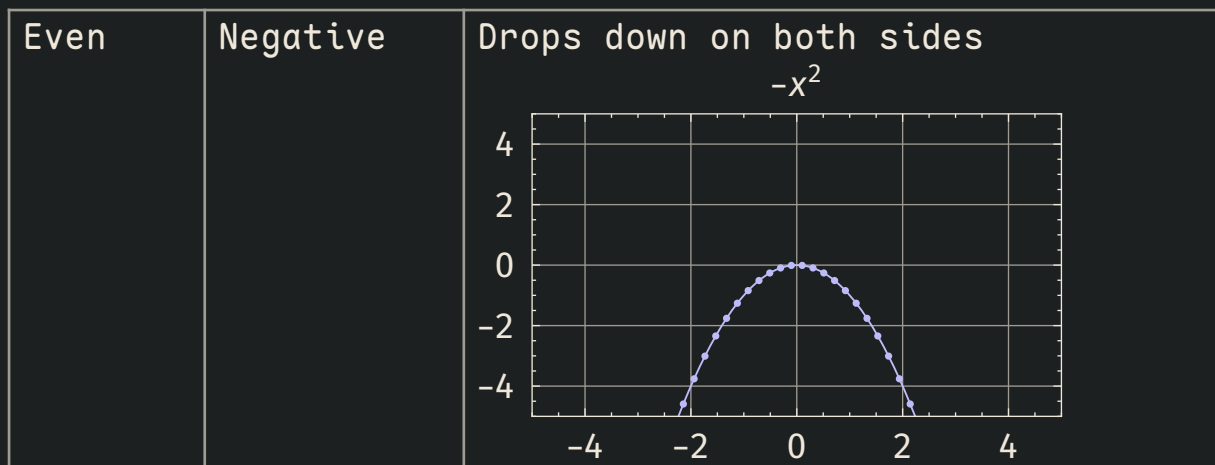
3.3. Inverse Trig Functions



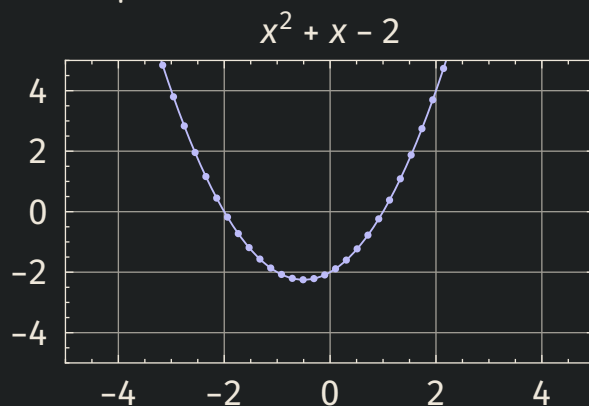
3.4. Polynomial Functions

- **Leading Coefficient Test:** The leading coefficient test helps us evaluate the end behaviour of a function. We look at the leading term in a polynomial, that is the term with the highest exponent and we look at its exponent and coefficient. We check if the exponent is even (x^2, x^4 , etc) or odd (x^3, x^5 etc) and if the coefficient is positive or negative.

Exponent	Coefficient	Behaviour
Odd	Positive	Drops down on the left, rises on the right x^3 
Odd	Negative	Rises on the left and drops down on the right $-x^3$ 
Even	Positive	Rises on both sides x^2 

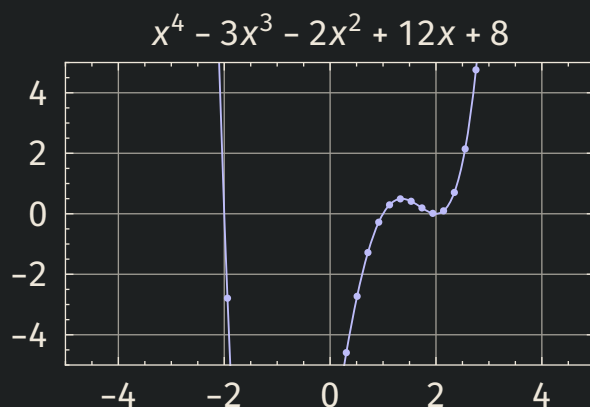


- Find the Zeroes: Find the values that makes the function equal to zero. This can be understood visually as x-intercepts. To do that we solve the function for x.
- If a zero is listed odd times then the graph will cross the x-axis at that value. Ex. For the polynomial $x^2 + x - 2$ the roots are $(x-1)(x+2)$. So, this will pass the x-axis at $x = -2, 1$.



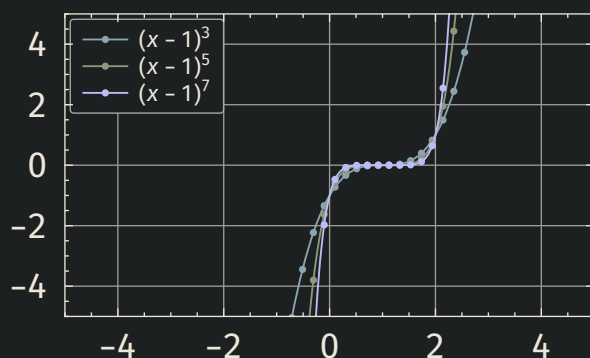
- If a zero is listed twice then the graph will touch the x-axis at that value. Ex. For the polynomial $x^4 - 3x^2 - 2x^2 + 12x + 8$ the roots are $(x+2)(x-1)(x-2)^2$. So, this will pass the x-axis at $x = -2, 1$ and touch the x-axis

at $x = 2$.



Note

As the value of the exponent in the factors increases, the graph tends to get less steep near that point. Ex.



3.5. Rational Functions

- Find the Asymptotes:
 - Vertical Asymptotes: The vertical asymptotes are the values that aren't in the domain. To find these we can set the denominator equal to zero and solve for it. Ex.

$$f(x) = \frac{x}{x^2 - 9}$$

$$\Rightarrow x^2 - 9 = 0$$

$$\Rightarrow (x + 3)(x - 3) = 0$$

$$\Rightarrow x = -3, 3$$

$\therefore$ The vertical asymptotes are, $x = -3, 3$

2. Horizontal Asymptotes: To find the horizontal asymptotes we have to take the numerator and denominator to their standard polynomial form $\left(\frac{ax^m}{bx^n}\right)$.

1. If $n > m$, $y = 0$ is a horizontal asymptote.

2. If $n < m$, no horizontal asymptote. But, there can be a slant asymptote.

3. If $n = m$, $y = \frac{a}{b}$ is a horizontal asymptote.

$$f(x) = \frac{x}{x^2 - 9}$$

$$\Rightarrow m = 1, n = 2$$

$\therefore y = 0$ is a Horizontal asymptote

3. Slant Asymptotes: If there is no horizontal asymptote ($m > n$), only then there can be a possibility of a slant. In the form $\frac{ax^m}{bx^n}$ when $m > n$, there can be a slant if and only if the degree of the numerator exceeds the degree of denominator by one ($m - n = 1$). To find the slant we need to do long division. Ex.

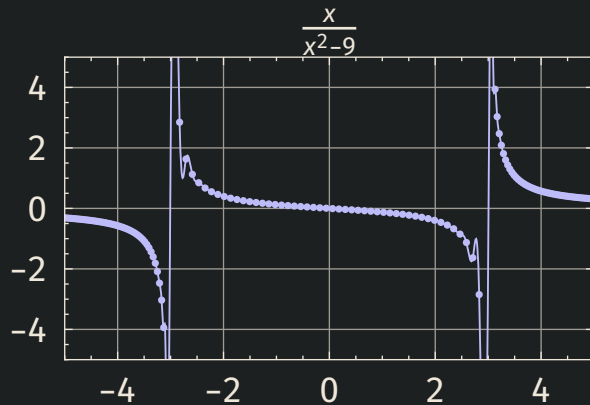
	$x + 2$
$x + 3$	$x^2 + 5x + 6$
	$-(x^2 + 3x + 0)$
	$0 + 2x + 6$
	$-(0 + 2x + 6)$
	0

$\therefore y = x + 2$ is a slant asymptote

Note

While a function can have multiple vertical asymptotes, it can only have a single horizontal or slant asymptote.

- Plot Points: Next, we need only to plot a few points to find the behaviour of the function as it approaches the asymptotes from each side.



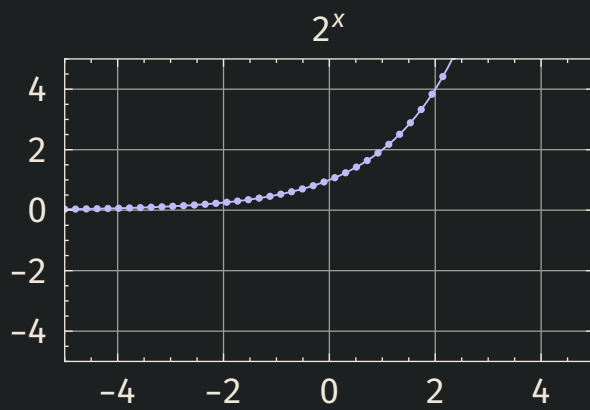
Note

If a factor $(x - c)$ cancels out completely from both the numerator and denominator, there is a hole at $x = c$, rather than a vertical asymptote. **Ex.**

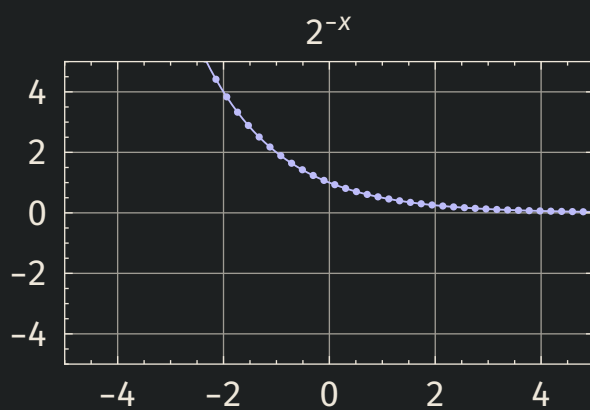
$$\begin{aligned}
 f(x) &= \frac{x^2 - 9}{x - 3} \\
 &= \frac{(x + 3)(x - 3)}{x - 3} \\
 &= x + 3 \\
 \therefore \text{There is a hole at } x = 3 \text{ or } (3, 6)
 \end{aligned}$$

3.6. Exponential Functions

- For the standard exponential form $y = a^x$,
 1. If $a > 1$ and x is positive or if $a < 1$ and x is negative, the function will rise upto the right.



1. If $a < 1$ and x is positive or if $a > 1$ and x is negative, the function will dip down to the right.



Note

In both cases, the domain is all real numbers $(-\infty, \infty)$ and range is all real numbers greater than zero $(0, \infty)$